

Mind the Long Tail: Understanding the Difficulty of Delay Detection in Business Processes (Supplementary Appendix)

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Abstract. This supplementary appendix provides additional methodological, experimental, and implementation details that complement the main paper. It expands on the experimental setup by documenting the hyperparameter tuning procedures, including the search spaces and optimization strategies used across all evaluated methods. Furthermore, it presents detailed descriptions of the data-level and algorithm-level approaches for imbalanced regression, including formal definitions, implementation adaptations, and extended discussions of their practical considerations.

The appendix also includes comprehensive empirical results that are not fully reported in the main paper, such as per-dataset performance tables, additional ablation studies, and analyses of smoothing techniques. These results offer a more granular view of model behavior across different target regions and datasets. Overall, this document aims to improve transparency, reproducibility, and interpretability of the experimental findings by providing the technical depth necessary to replicate and extend the work.

1 Introduction

This appendix supplements the main paper “Mind the Long Tail: Understanding the Difficulty of Delay Detection in Business Processes” by providing detailed methodological descriptions, implementation specifics, and extended experimental results. While the main paper focuses on the conceptual insights and high-level empirical findings, this document offers the technical depth required for full reproducibility and a more comprehensive understanding of the evaluated approaches.

First, we describe the hyperparameter tuning procedure used throughout the study, including the Bayesian optimization framework, search spaces, and method-specific configurations. This is intended to clarify how model configurations were selected and to ensure that the reported results can be reliably reproduced.

Second, we provide a detailed account of the data-level and algorithm-level approaches for imbalanced regression. This includes formal definitions of the methods, implementation details tailored to the predictive process monitoring setting, and clarifications on how these techniques interact with sequential event

Table 1: Hyperparameter search space for each method. Fixed values indicate no tuning.

Method	Hyperparameter	Search space
All	Learning rate	$[10^{-5}, 10^{-2}]$ (log)
Vanilla	Loss	MAE (fixed)
CSW	Loss	{MAE, MSE, Huber}
	Reweighting	{inverse, sqrt_inv}
	Bins	{10, 20, 50}
EAL	Loss	{Focal-MAE, Focal-MSE}
	β	$[10^{-2}, 1]$ (log)
	γ	$[0.5, 5]$
BMSE	Loss	BMSE (fixed)
SERA	Extreme type	{high, both}
	Asymmetry	{False, True}
SMOBN	k	(default = 5)
	Relevance threshold	(default = 0.5)
	Over-sampling ratio	(default = 10)
Survival	Bins	{10, 15, 20}
	Binning	{quantile, hybrid_tail}
	Tail fraction	{0.1, 0.2, 0.3}
	Tail bin fraction	{0.3, 0.4, 0.5}
	Prediction type	{mean, median}

data. In particular, we elaborate on loss reweighting strategies, distribution-aware objectives, and relevance-based methods, as well as their extensions through feature and label distribution smoothing.

Third, the appendix presents extended experimental results that complement those in the main paper. These include per-dataset performance tables, additional comparisons across methods, and analyses of the impact of resampling and smoothing techniques. The detailed results provide further insight into how different approaches behave across target-frequency regions and highlight the variability of performance across datasets.

Overall, this appendix is intended as a technical companion to the main paper, offering the additional detail necessary for replication, deeper analysis, and future research building upon this work.

2 Hyperparameter Tuning

We tune hyperparameters using Bayesian optimization implemented via Ax (Add reference to the library). Each configuration is evaluated on the validation set, and the objective is to minimize validation loss. The optimization starts with Sobol sampling followed by Bayesian optimization. The search space is defined per method. Common parameters include the learning rate (log-scale) and loss-specific settings, while method-specific parameters are activated only when relevant. The number of trials is 10 for Vanilla, and 45 for all other approaches. Table 1 summarizes the tuned hyperparameters and their search spaces.

Table 2: Performance difference between the LSTM trained with SMOGN and the same model trained on the original (imbalanced) data ($\Delta = \text{SMOGN} - \text{Original}$). Negative values indicate an improvement (lower nMAE).

Log	All	Many	Med.	Few
P2P	-0.06	0.03	0.00	-0.18
BPIC17W	-0.01	0.05	-0.06	-0.16
BPIC15-1	0.16	0.17	0.22	-0.39
BPIC15-2	1.34	1.33	1.51	-1.85
BPIC15-3	0.46	0.42	0.87	-1.20
BPIC15-4	0.05	0.05	0.02	-0.01
BPIC15-5	0.07	0.04	0.14	0.20
Helpdesk	-0.01	0.01	-0.08	-0.11
Sepsis	0.05	0.01	0.13	0.14
BPIC20ID	0.03	0.04	0.00	-0.06
BPIC20DD	0.00	0.07	-0.05	-0.16
BPIC20PTC	0.04	0.08	-0.02	-0.33
BPIC20TPD	0.05	0.13	-0.07	-0.05
BPIC20RFP	0.01	0.08	-0.07	-0.07
Average	0.16 (+17.5%)	0.18 (+31.4%)	0.18 (+20.5%)	-0.30 (-8.0%)

3 Data-level Approaches

This section provides additional details on SMOGN as a data-level approach for handling imbalanced regression. We report the specific configurations and hyperparameters employed in our study, as summarized in Table 3.

4 Algorithm-level Approaches

Algorithm-level approaches mitigate imbalance by modifying the regression objective to assign greater weight to underrepresented target regions. They can also be combined with feature and label distribution smoothing techniques as we explain in this section. We first evaluate the performance of algorithm-level approaches and then evaluate their combination with feature and label smoothing.

4.1 Reweighting the Loss Function

We consider cost-sensitive weighting, distribution-balanced objectives, error-aware losses, and relevance-based error measures.

Cost-Sensitive Weighting (CSW). The empirical density of $p_{\text{train}}(y)$ can be used to assign static loss weights to different target ranges. In practice, this is done by discretizing the target domain into I consecutive bins of equal width and mapping each target value to a bin index $i \in \{1, \dots, I\}$. Let n_i denote the

number of training samples in bin i and N the total number of samples. The empirical bin frequency is $\hat{p}_i = n_i/N$. The loss weight in Eq. ?? for samples in bin i is then set to

$$w_i = \frac{1}{\hat{p}_i} \quad \text{or} \quad w_i = \frac{1}{\sqrt{\hat{p}_i}} \quad (1)$$

thereby down-weighting overrepresented regions and up-weighting underrepresented ones [6].

Balanced Mean Squared Error (BMSE). BMSE extends static bin-based weighting by incorporating the training target distribution $p_{\text{train}}(y)$ directly into the regression objective. We adopt the batch-based Monte Carlo (BMC) variant of BMSE [2], which treats the target values in a mini-batch $B_y = \{y_{(1)}, \dots, y_{(n)}\}$ as samples from $p_{\text{train}}(y)$ and uses them to normalize prediction errors within the batch. For a prediction $\hat{y}$ and target y , the BMC loss is defined as:

$$\mathcal{L}_{\text{BMC}} = -\log \frac{\exp\left(-\frac{\|\hat{y}-y\|^2}{2\sigma_{\text{noise}}^2}\right)}{\sum_{y' \in B_y} \exp\left(-\frac{\|\hat{y}-y'\|^2}{2\sigma_{\text{noise}}^2}\right)} \quad (2)$$

This objective can be interpreted as a softmax over squared distances within the batch: the numerator favors small error for the true target, while the denominator contrasts it against errors to other batch targets. As a result, prediction errors in densely populated target regions contribute less relative weight than errors in sparse regions. The temperature parameter $\tau = 2\sigma_{\text{noise}}^2$ controls the sharpness of this contrast. Following [2], we treat σ_{noise} as a learnable parameter and optimize it jointly with the model parameters during training.

Error-Aware Loss (EAL). Error-aware loss functions adapt sample weights based on the magnitude of the model’s prediction error. For instance, Focal-R [6] loss emphasizes hard examples by down-weighting those with small errors through a sigmoid-based scaling factor. It is defined as

$$\mathcal{L}_{\text{Focal-R}} = \frac{1}{N} \sum_{i=1}^N \sigma(|\beta e_i|)^\gamma e_i, \quad (3)$$

where e_i denotes the absolute or squared error of the i -th sample, $\sigma(\cdot)$ is the sigmoid function, and β and γ are hyperparameters controlling the scaling and focusing strength. Larger errors therefore receive relatively higher weights, increasing the contribution of difficult examples during optimization.

Squared Error Relevance Area (SERA). SERA is a relevance-based criterion for imbalanced regression that can be used both as a training objective and as an evaluation metric [4,3]. It relies on a relevance function $\phi : Y \rightarrow [0, 1]$ that assigns each target value an importance score, enabling non-uniform preference over the continuous target domain [5,1]. In this work, ϕ is automatically derived from the training target distribution using box-plot statistics, as proposed by Ribeiro and Moniz [3]. This yields higher relevance scores for extreme remaining time values.

Given a relevance threshold r , the set of relevant samples is defined as $\mathcal{D}^r = \{(\mathbf{x}, y) \in \mathcal{D} : \phi(y) \geq r\}$. SERA aggregates squared errors over relevance-defined subsets across all thresholds:

$$\mathcal{L}_{\text{SERA}} = \int_0^1 \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}^r} (y_i - \hat{y}_i)^2 dr. \quad (4)$$

Errors on high-relevance targets are counted across a wider range of thresholds and therefore receive greater overall weight, while low-relevance targets contribute less. This mechanism emphasizes extreme values without fully discarding the rest of the target range [4].

4.2 Feature and Label Smoothing

Algorithm-level imbalanced regression approaches can also be combined with feature and label distribution smoothing techniques that exploit the continuity of the target space.

Label Distribution Smoothing. In imbalanced regression, simply counting how often each target value appears in the training data can be misleading. Unlike classification, target values are continuous and nearby values are typically similar. As a result, prediction difficulty does not necessarily follow the raw label frequency. Label Distribution Smoothing (LDS) addresses this issue by estimating an *effective* label density that also accounts for neighboring target values. Concretely, the continuous target space is first discretized into bins, as in cost-sensitive weighting. Instead of directly using the bin counts, LDS smooths them by spreading each bin’s mass to its neighbors using a symmetric kernel. This produces a smoothed label distribution in which nearby target values support each other. Bins located in sparse regions but surrounded by populated neighbors are therefore treated as less rare than their raw counts suggest. The resulting smoothed density can then replace the empirical distribution in standard loss re-weighting schemes, yielding more stable and more realistic weights [6].

Feature Distribution Smoothing (FDS). Under skewed target distributions, the encoder $\mathbf{z} = g(\mathbf{x}, \theta)$ tends to learn latent features that are dominated by well-represented target regions. FDS reduces this effect by smoothing feature statistics across nearby target values. After the encoder produces latent features, the target space is discretized into bins and the mean and covariance of latent features within each bin are estimated. FDS smooths these statistics across neighboring bins using a symmetric kernel, assuming that nearby target values should have similar latent representations. The resulting features $\tilde{\mathbf{z}}$ are then passed to the prediction head: $\hat{y} = h(\tilde{\mathbf{z}}, \psi)$. Implemented as a lightweight feature calibration layer before the prediction head, FDS can be integrated into any neural network, complementing previous algorithm-level approaches [6].

Experiments. We next evaluate whether combining feature and label smoothing with imbalance-aware objectives improves prediction performance. Note that LDS can only be combined with approaches that rely on sample reweighting.

Table 3: Results DALSTM

Log	IR	nMAE ↓			
		All	Many	Med.	Few
P2P	Vanilla	0.88 (0.01)	0.14 (0.01)	0.21 (0.00)	1.83 (0.02)
	SMOBN	0.82 (0.02)	0.17 (0.01)	0.21 (0.02)	1.65 (0.05)
	CSW	0.88 (0.07)	0.29 (0.07)	0.18 (0.05)	1.70 (0.13)
	EAL	0.83 (0.01)	0.24 (0.04)	0.26 (0.06)	1.59 (0.06)
	BMSE	0.84 (0.03)	0.30 (0.05)	0.33 (0.08)	1.53 (0.05)
	SERA	0.93 (0.12)	0.94 (0.32)	0.59 (0.13)	1.07 (0.05)
	Survival	0.84 (0.03)	0.47 (0.09)	0.51 (0.13)	1.31 (0.08)
BPI17W	Vanilla	0.89 (0.00)	0.50 (0.01)	0.78 (0.01)	3.75 (0.02)
	SMOBN	0.88 (0.00)	0.55 (0.02)	0.72 (0.02)	3.59 (0.02)
	CSW	0.89 (0.01)	0.51 (0.01)	0.77 (0.02)	3.75 (0.03)
	EAL	0.89 (0.00)	0.50 (0.01)	0.78 (0.01)	3.75 (0.02)
	BMSE	1.01 (0.02)	0.87 (0.04)	0.58 (0.01)	3.28 (0.04)
	SERA	3.88 (0.13)	4.68 (0.17)	3.04 (0.09)	1.20 (0.03)
	Survival	1.54 (0.01)	1.52 (0.01)	1.44 (0.02)	1.96 (0.01)
BPI15-1	Vanilla	0.99 (0.07)	0.88 (0.09)	0.87 (0.09)	4.90 (0.35)
	SMOBN	1.15 (0.09)	1.05 (0.09)	1.09 (0.12)	4.51 (0.15)
	CSW	0.89 (0.01)	0.76 (0.04)	0.82 (0.05)	5.02 (0.14)
	EAL	0.87 (0.04)	0.70 (0.07)	0.88 (0.07)	5.03 (0.07)
	BMSE	1.33 (0.13)	1.34 (0.18)	1.08 (0.18)	4.10 (0.12)
	SERA	5.52 (0.62)	6.12 (0.78)	4.63 (0.39)	1.51 (0.40)
	Survival	1.69 (0.35)	1.54 (0.43)	1.83 (0.25)	3.77 (0.28)
BPI15-2	Vanilla	1.12 (0.12)	1.14 (0.15)	0.93 (0.11)	4.04 (0.33)
	SMOBN	2.46 (0.44)	2.47 (0.53)	2.44 (0.10)	2.19 (0.23)
	CSW	1.11 (0.19)	1.06 (0.24)	1.27 (0.17)	3.32 (0.47)
	EAL	1.25 (0.19)	1.30 (0.21)	0.98 (0.23)	3.83 (0.27)
	BMSE	1.36 (0.35)	1.45 (0.43)	0.93 (0.14)	3.80 (0.51)
	SERA	6.43 (0.00)	6.91 (0.00)	4.57 (0.00)	1.22 (0.00)
	Survival	2.94 (0.13)	3.07 (0.17)	2.42 (0.10)	2.40 (0.26)
BPI15-3	Vanilla	0.88 (0.03)	0.41 (0.06)	1.06 (0.01)	5.11 (0.13)
	SMOBN	1.34 (0.11)	0.83 (0.14)	1.93 (0.14)	3.91 (0.11)
	CSW	0.86 (0.06)	0.39 (0.05)	1.03 (0.06)	4.92 (0.35)
	EAL	0.90 (0.03)	0.42 (0.04)	1.11 (0.02)	4.95 (0.09)
	BMSE	1.04 (0.09)	0.61 (0.10)	1.26 (0.13)	4.57 (0.21)
	SERA	4.50 (0.13)	5.00 (0.18)	3.94 (0.08)	1.95 (0.10)
	Survival	1.35 (0.08)	0.85 (0.07)	2.09 (0.13)	3.14 (0.14)
BPI15-4	Vanilla	1.09 (0.02)	1.12 (0.03)	0.45 (0.03)	1.66 (0.04)
	SMOBN	1.14 (0.09)	1.17 (0.11)	0.47 (0.12)	1.65 (0.13)
	CSW	1.09 (0.03)	1.12 (0.03)	0.42 (0.03)	1.64 (0.05)
	EAL	1.08 (0.04)	1.10 (0.04)	0.45 (0.03)	1.67 (0.06)
	BMSE	1.26 (0.03)	1.31 (0.04)	0.30 (0.03)	1.38 (0.08)
	SERA	2.84 (0.00)	2.95 (0.00)	1.31 (0.00)	0.26 (0.00)
	Survival	1.67 (0.08)	1.73 (0.09)	0.82 (0.08)	0.92 (0.12)
BPI15-5	Vanilla	0.81 (0.03)	0.51 (0.05)	1.11 (0.06)	3.68 (0.14)
	SMOBN	0.88 (0.03)	0.55 (0.03)	1.25 (0.18)	3.88 (0.20)
	CSW	0.83 (0.02)	0.56 (0.06)	1.08 (0.16)	3.61 (0.11)
	EAL	0.83 (0.01)	0.56 (0.06)	1.05 (0.13)	3.48 (0.10)
	BMSE	1.06 (0.03)	1.04 (0.06)	0.61 (0.05)	3.25 (0.06)
	SERA	6.08 (0.11)	6.70 (0.12)	4.87 (0.12)	2.51 (0.19)
	Survival	1.18 (0.11)	1.03 (0.12)	1.37 (0.28)	2.57 (0.25)

Table 4: Results DALSTM

Log	IR	nMAE ↓			
		All	Many	Med.	Few
Helpdesk	Vanilla	0.87 (0.01)	0.87 (0.01)	0.62 (0.01)	1.44 (0.01)
	SMOBN	0.86 (0.02)	0.88 (0.03)	0.54 (0.04)	1.33 (0.03)
	CSW	0.91 (0.01)	0.97 (0.01)	0.46 (0.01)	1.25 (0.01)
	EAL	0.91 (0.01)	0.96 (0.01)	0.47 (0.01)	1.27 (0.01)
	BMSE	0.91 (0.02)	0.96 (0.03)	0.47 (0.02)	1.27 (0.02)
	SERA	1.82 (0.02)	2.17 (0.02)	0.61 (0.01)	0.31 (0.00)
	Survival	0.88 (0.02)	0.93 (0.03)	0.48 (0.02)	1.27 (0.03)
Sepsis	Vanilla	0.94 (0.00)	0.27 (0.02)	2.03 (0.03)	7.76 (0.04)
	SMOBN	0.99 (0.00)	0.28 (0.00)	2.16 (0.01)	7.90 (0.01)
	CSW	0.94 (0.00)	0.23 (0.01)	2.14 (0.03)	7.86 (0.02)
	EAL	0.94 (0.00)	0.23 (0.01)	2.12 (0.03)	7.85 (0.02)
	BMSE	1.71 (0.10)	1.62 (0.14)	1.32 (0.02)	6.36 (0.12)
	SERA	6.91 (0.02)	7.70 (0.02)	5.34 (0.02)	0.88 (0.01)
	Survival	2.21 (0.33)	2.18 (0.41)	1.99 (0.27)	4.58 (0.36)
BPIC20ID	Vanilla	0.72 (0.01)	0.44 (0.00)	1.13 (0.01)	5.05 (0.03)
	SMOBN	0.75 (0.01)	0.48 (0.00)	1.13 (0.02)	4.99 (0.03)
	CSW	0.72 (0.01)	0.44 (0.00)	1.12 (0.02)	5.07 (0.04)
	EAL	0.72 (0.00)	0.45 (0.00)	1.11 (0.00)	5.06 (0.02)
	BMSE	0.83 (0.01)	0.58 (0.03)	1.19 (0.02)	4.67 (0.02)
	SERA	5.24 (0.02)	6.01 (0.02)	3.82 (0.02)	2.16 (0.00)
	Survival	1.32 (0.04)	0.79 (0.02)	2.28 (0.10)	3.61 (0.12)
BPIC20DD	Vanilla	0.87 (0.00)	0.38 (0.01)	0.62 (0.01)	3.34 (0.02)
	SMOBN	0.87 (0.00)	0.45 (0.01)	0.57 (0.01)	3.18 (0.03)
	CSW	0.87 (0.00)	0.40 (0.01)	0.60 (0.01)	3.30 (0.02)
	EAL	0.87 (0.00)	0.42 (0.01)	0.58 (0.01)	3.25 (0.02)
	BMSE	0.94 (0.01)	0.68 (0.02)	0.56 (0.00)	2.72 (0.02)
	SERA	2.71 (0.03)	3.35 (0.04)	2.10 (0.04)	1.43 (0.01)
	Survival	1.42 (0.07)	1.26 (0.06)	1.48 (0.14)	1.94 (0.03)
BPIC20PTC	Vanilla	0.79 (0.01)	0.48 (0.01)	1.08 (0.01)	4.59 (0.09)
	SMOBN	0.83 (0.00)	0.56 (0.01)	1.06 (0.01)	4.26 (0.04)
	CSW	0.98 (0.01)	0.73 (0.01)	1.28 (0.03)	3.37 (0.06)
	EAL	0.83 (0.00)	0.55 (0.01)	1.09 (0.01)	4.26 (0.02)
	BMSE	1.00 (0.01)	0.74 (0.01)	1.30 (0.02)	3.31 (0.05)
	SERA	2.92 (0.10)	2.64 (0.13)	3.65 (0.07)	0.98 (0.01)
	Survival	1.71 (0.04)	1.12 (0.04)	2.95 (0.04)	1.08 (0.04)
BPIC20TPD	Vanilla	0.76 (0.01)	0.49 (0.02)	0.86 (0.01)	2.99 (0.09)
	SMOBN	0.81 (0.01)	0.62 (0.01)	0.79 (0.02)	2.94 (0.06)
	CSW	0.76 (0.01)	0.50 (0.01)	0.85 (0.02)	2.96 (0.06)
	EAL	0.76 (0.00)	0.50 (0.01)	0.85 (0.01)	2.96 (0.04)
	BMSE	0.88 (0.02)	0.71 (0.02)	0.89 (0.04)	2.53 (0.08)
	SERA	3.34 (0.01)	4.09 (0.01)	2.30 (0.01)	1.06 (0.00)
	Survival	1.16 (0.03)	1.03 (0.04)	1.34 (0.03)	1.58 (0.06)
BPIC20RFP	Vanilla	0.87 (0.00)	0.36 (0.01)	0.64 (0.01)	2.99 (0.01)
	SMOBN	0.88 (0.00)	0.44 (0.01)	0.57 (0.01)	2.92 (0.02)
	CSW	0.87 (0.00)	0.38 (0.01)	0.62 (0.01)	2.96 (0.02)
	EAL	0.87 (0.00)	0.42 (0.01)	0.59 (0.01)	2.85 (0.01)
	BMSE	0.92 (0.02)	0.65 (0.08)	0.56 (0.01)	2.45 (0.13)
	SERA	2.91 (0.03)	3.59 (0.03)	2.56 (0.04)	1.22 (0.01)
	Survival	1.32 (0.03)	1.23 (0.03)	1.47 (0.05)	1.38 (0.02)

Since SERA employs a relevance-weighted loss and BMSE uses a pairwise probabilistic objective that does not incorporate sample weights, LDS is not directly compatible with these approaches.

Table 5 reports the relative change in nMAE after combining algorithm-level imbalanced regression methods with feature and label smoothing. Overall, smoothing produces only minor and inconsistent effects, with most Δ values being positive, indicating increased prediction error compared to the corresponding baseline models. For CSW, EAL, and BMSE, feature distribution smoothing results in negligible changes and no statistically significant improvements. When combined with SERA, FDS yields significant improvements in the overall and many-shot regions (-7.97% and -9.05%, $p = 0.011$), but does not improve performance in the few-shot region. Label distribution smoothing exhibits unstable behavior: while it slightly improves the few-shot region for EAL (-3.31%, $p = 0.013$), it substantially degrades performance in the many-shot region (+11.71%, $p = 0.035$). Overall, smoothing techniques frequently increase prediction errors and do not consistently improve performance, particularly in the few-shot region corresponding to extreme remaining times. Statistical significance is assessed using a paired Wilcoxon signed-rank test across datasets, where p -values below 0.05 indicate a consistent difference between smoothed and baseline models. These findings further support the observation that the challenge of predicting extreme remaining times cannot be addressed solely through imbalance-aware objectives or distribution smoothing techniques.

Table 5: Average relative change in nMAE (%) after applying smoothing to imbalanced regression approaches. Negative values indicate improvement. Statistical significance is assessed with a paired Wilcoxon signed-rank test across datasets.

IR	Smoothing	All		Many		Medium		Few	
		Δ	p	Δ	p	Δ	p	Δ	p
CSW	FDS	-0.14%	0.715	-1.11%	0.426	+5.96%	0.296	+3.45%	0.761
EAL	FDS	-0.55%	1.000	-1.99%	0.296	+5.49%	0.217	+0.88%	0.715
BMSE	FDS	-0.95%	0.091	-1.64%	0.296	+7.11%	0.808	+2.88%	0.391
SERA	FDS	-7.97%	0.011	-9.05%	0.011	-1.01%	0.194	+0.70%	0.626
CSW	LDS	+0.11%	0.391	+0.85%	0.502	+4.73%	0.761	+3.26%	0.670
EAL	LDS	+2.85%	0.078	+11.71%	0.035	+8.36%	0.391	-3.31%	0.013
CSW	LDS+FDS	-1.55%	0.391	-2.87%	0.326	+7.64%	0.358	+6.43%	0.326
EAL	LDS+FDS	+2.32%	0.761	+4.25%	0.670	+12.86%	0.135	+0.90%	0.903

5 Delay Detection via Uncertainty Modeling

This section complements the results presented in the main paper by providing additional evaluation of the uncertainty-aware delay detection approach under a stricter delay definition ($q = 0.9$). While the main paper demonstrates

Table 6: Delay detection performance (threshold $q = 0.9$). Average Recall, Precision, F1-score, and PR-AUC over five random seeds for baseline and uncertainty-aware models.

Dataset	Baseline Classifier				Uncertainty-Aware Model			
	Recall	Precision	F1-Score	PRAUC	Recall	Precision	F1-Score	PRAUC
P2P	0.16	1.00	0.27	0.66	0.85	0.90	0.87	0.85
BPIC17W	0.19	0.29	0.23	0.20	0.59	0.55	0.57	0.40
BPIC15-1	0.31	0.07	0.11	0.04	0.63	0.09	0.15	0.06
BPIC15-2		0.00	0.00			0.00	0.00	
BPIC15-3	0.25	0.31	0.27	0.12	0.70	0.50	0.58	0.37
BPIC15-4	0.00	0.00	0.00	0.01	0.33	0.01	0.02	0.01
BPIC15-5	0.28	0.34	0.30	0.14	0.52	0.21	0.29	0.14
HelpDesk	0.00	0.11	0.00	0.11	0.58	0.17	0.26	0.14
Sepsis	0.04	0.04	0.04	0.04	0.56	0.06	0.11	0.05
BPIC20ID	0.00	0.00	0.00	0.00	0.22	0.02	0.03	0.01
BPIC20DD	0.01	0.33	0.02	0.14	0.46	0.57	0.51	0.34
BPIC20PTC	0.00		0.00	0.01	0.72	0.04	0.08	0.03
BPIC20TPD	0.00	0.00	0.00	0.03	0.44	0.15	0.23	0.09
BPIC20RFP	0.05	0.83	0.09	0.21	0.56	0.59	0.57	0.40

that predictive uncertainty provides actionable signals for identifying high-risk cases, here we report the corresponding detailed performance metrics across all datasets. Table 6 summarizes these results, confirming that the benefits of uncertainty modeling persist under more stringent delay thresholds and further supporting its practical utility for prioritizing delayed cases.

References

1. Branco, P., Ribeiro, R.P., Torgo, L.: Ubl: an r package for utility-based learning. arXiv preprint arXiv:1604.08079 (2016)
2. Ren, J., Zhang, M., Yu, C., Liu, Z.: Balanced mse for imbalanced visual regression. In: CVPR. pp. 7926–7935 (2022)
3. Ribeiro, R.P., Moniz, N.: Imbalanced regression and extreme value prediction. Machine Learning **109**(9), 1803–1835 (2020)
4. Silva, A., Ribeiro, R.P., Moniz, N.: Model optimization in imbalanced regression. In: International Conference on Discovery Science. pp. 3–21. Springer (2022)
5. Torgo, L., Ribeiro, R.: Utility-based regression. In: European conference on principles of data mining and knowledge discovery. pp. 597–604. Springer (2007)
6. Yang, Y., Zha, K., Chen, Y., Wang, H., Katabi, D.: Delving into deep imbalanced regression. In: ICML. pp. 11842–11851. PMLR (2021)